

The background of the slide features a series of overlapping, translucent, wavy bands in shades of blue and green, creating a sense of depth and movement. These bands are set against a solid dark grey background.

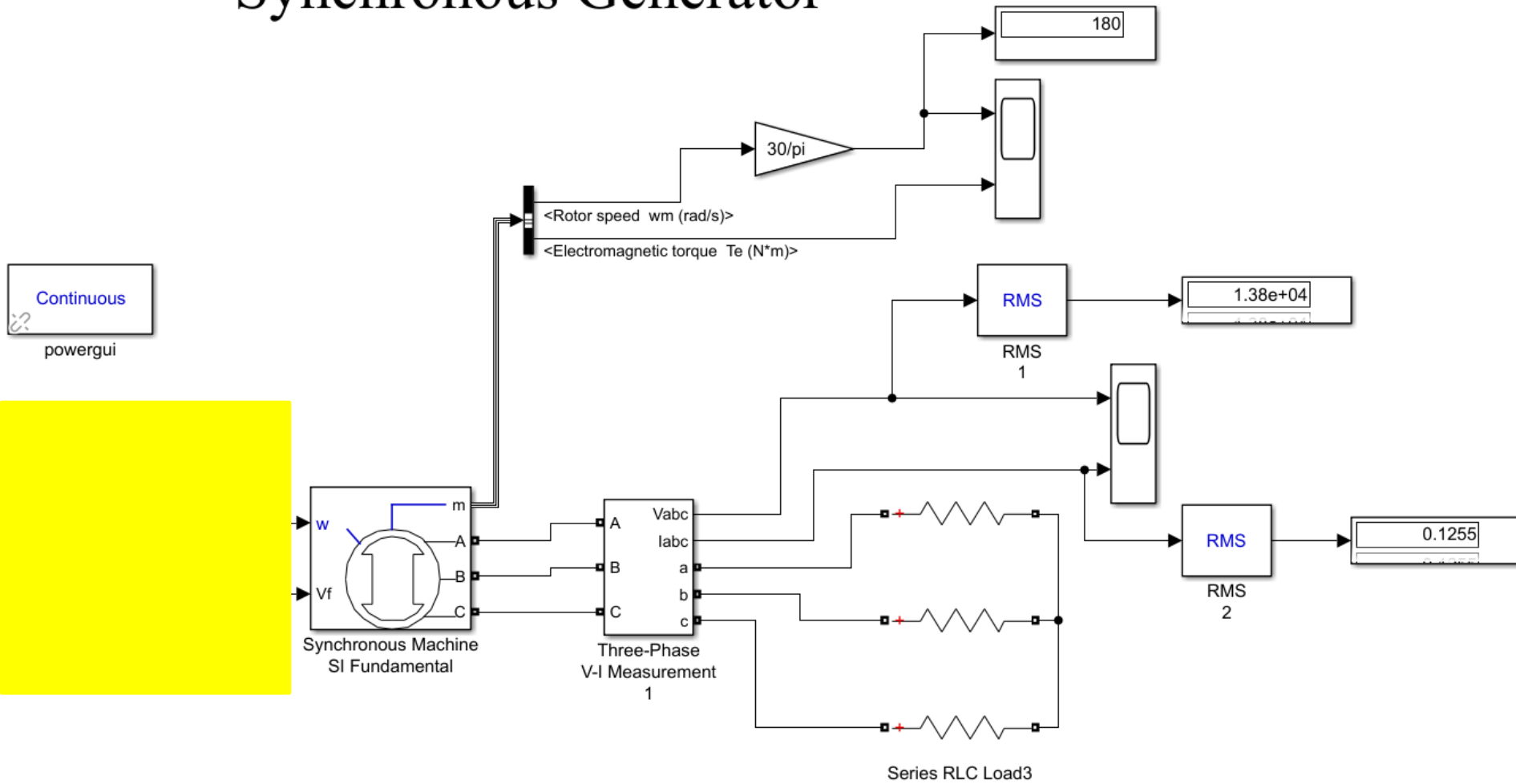
Section 4

Synchronous Generator

Simulation **Test 3**



Synchronous Generator



Parameters

Block Parameters: Synchronous Machine SI Fundamental

Synchronous Machine (mask) (link)

Implements a 3-phase synchronous machine modelled in the dq rotor reference frame. Stator windings are connected in wye to an internal neutral point.

Configuration Parameters Load Flow

Nominal power, voltage, frequency, field current [Pn(VA) Vn(Vrms) fn(Hz) ifn(A)]: [187E6 13800 60 1087]

Stator [Rs(ohm) Lj,Lmd,Lmq(H) (Lc(H))]: [2.9069E-03 3.0892E-04 3.2164E-03 9.7153E-04]

Field [Rf(ohm) Lfld(H)]: [5.9013E-04 3.0712E-04]

Dampers [Rkd',Llkd' Rkq1',Llkq1'] (R=ohm,L=H): [1.1900E-02 4.9076E-04 2.0081E-02 1.0365E-03]

Pole pairs: 20

Initial conditions [dw(%) th(deg) ia,ib,ic(A) pha,phb,phc(deg) Vf(V)]: [0 0 0 0 0 0 0 0 70.3192]

☐ Simulate saturation Plot

[ifd(A) ; Vt(VLL rms)]: [1,1175.9,1293.6,1430.2,1583.7;9660,10623,12243,13063,13757,14437,15180,15890,16567]

OK Cancel Help Apply

Block Parameters: Series RLC Load2

Series RLC Load (mask) (link)

Implements a series RLC load.

Parameters Load Flow

Nominal voltage Vn (Vrms): 13800/sqrt(3)

Nominal frequency fn (Hz): 60

Active power P (W): 1000

Inductive reactive power QL (positive var): 0

Capacitive reactive power Qc (negative var): 0

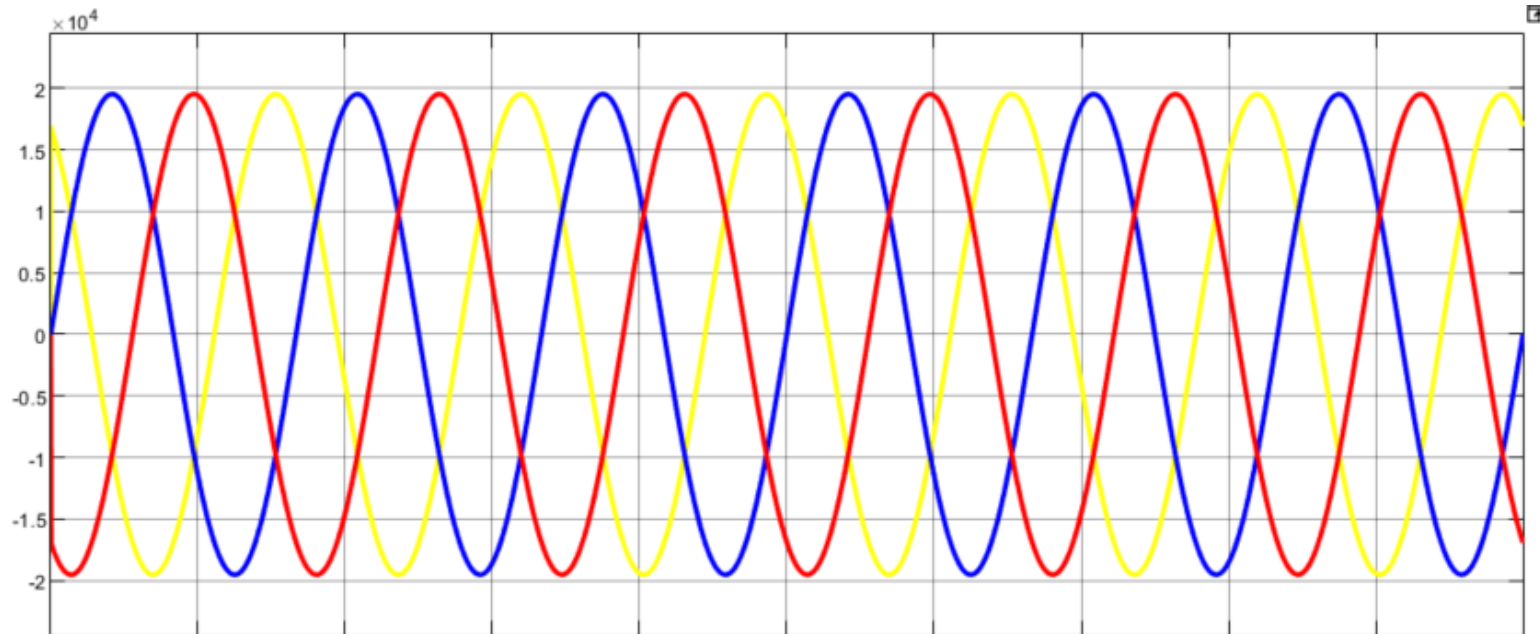
☐ Set the initial capacitor voltage

☐ Set the initial inductor current

Measurements None

OK Cancel Help Apply

Voltage:



Current:

